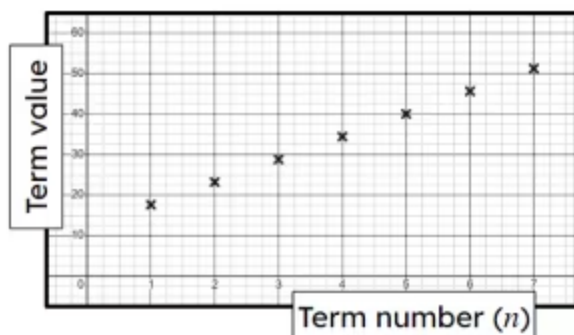


Graphing special number sequences using technology

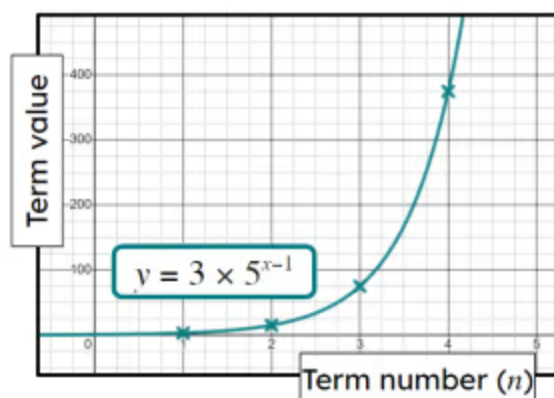
- 1 The sequence with n^{th} term $5.6n + 12$ is graphed in the diagram. Is 40 a term in the sequence? Tick **1** correct answer



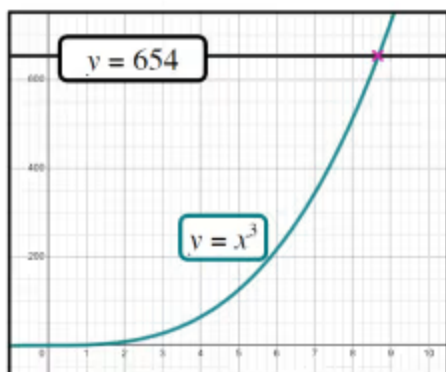
☒ yes

☐ no

- 2 The 4th term in the sequence $3 \times 5^{n-1}$ is 375. Use Desmos or the diagram to help you. Fill in the blank



- 3 Is the number 654 in the sequence with n^{th} term rule n^3 ? Use Desmos or the diagram to help you. Tick **1** correct answer



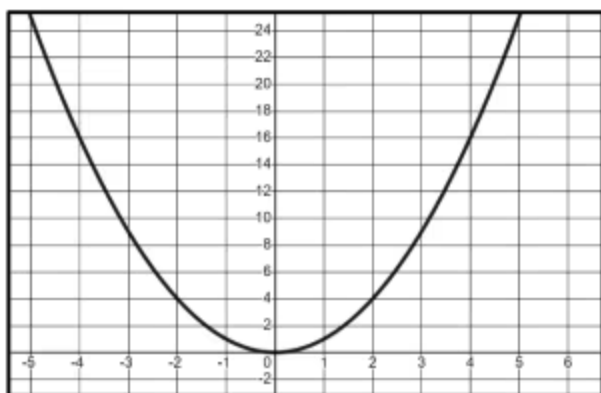
☐ Yes, because the two graphs intersect.

☐ Yes, because $\sqrt[3]{654}$ is an integer.

☒ No, because the graphs do not intersect at an integer x value.

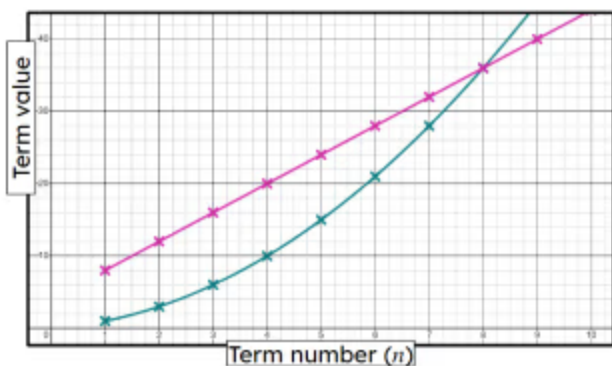
☐ No, because 654 is not on the curve with equation $y = n^3$.

- 4 Why does this diagram not represent the sequence of all square numbers in ascending order? Tick 2 correct answers



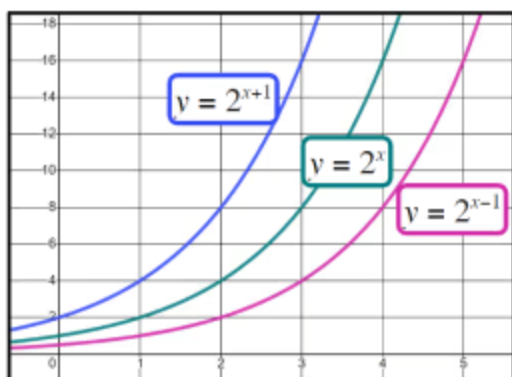
- ☒ The sequence has integer term numbers so the curve should not be drawn.
- ☐ This should be a linear graph not a curve.
- ☐ The y axis scale is double the scale on the x axis.
- ☐ The square numbers are not on the curve.
- ☒ Sequences start when n is 1 so there should not be coordinates when $x < 1$.

- 5 This graph shows the sequence of triangular numbers and the sequence $4n + 4$ (the line and curve are drawn to help you). Which of these statements are true? Tick 2 correct answers



- ☐ The number 1 is in both sequences.
- ☐ The number 20 is in both sequences.
- ☒ The number 28 is in both sequences.
- ☐ The 6th term in both sequences is the same.
- ☒ The 8th term in both sequences are the same.

- 6 Which is the correct n^{th} term rule for the geometric sequence which starts 1, 2, 4, 8, 16, ...? Tick 1 correct answer



- ☐ 2^{n+1}
- ☐ 2^n
- ☒ $2^n - 1$



